Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Medicinal plants of Sri Lanka are an important aspect of its rich biodiversity and have made a significant impact on the development of Ayurveda medicine, which is a traditional system of

medicine used in Sri Lanka for over 3000 years. There are about 1400 type of hurbal plants found Ceylon, and most of them are commonly used in traditional medicine and herbal remedies. However, the traditional knowledge about the medicinal plants is gradually declining as the traditional practitioners are aging and the transfer of knowledge to the younger generation is limited. As a result, today people are not able to identify the medicinal plants and their medicinal benefits. Moreover, most of the digital platforms require technical knowledge and/or do not allow plant identification.

The proposal of this study is to establish a concept of developing an Ayurvedic Plant Detection System that will help users recognize Sri Lankan medicinal plants through a scanning system. The Ayurvedic Plant Detection System will allow a user to scan a picture of a Sri Lankan medicinal plant's leaves or flowers using a smartphone's camera. The picture will then be analyzed using artificial intelligence and machine learning techniques, particularly image recognition technology, to recognize the type of plant. After recognizing the type of plant, it will then retrieve relevant information from a database of medicinal plants that contain information such as the name of the plant, description, benefits, parts used, and Ayurvedic applications.

The proposed solution can be implemented mobile app in order to provide easy access for the public, students, and researchers. The app can be developed in such a manner that the backend system can process the plant image and provide the required information from the plant database, while the app can provide the required functionality for the public to access the system easily. This project can be helpful in creating awareness among the public regarding the medicinal plants in Sri Lanka, and the traditional knowledge of the herbal plants can be retained through this system. This system can also help in achieving the Sustainable Development Goals such as Good Health and Well-being (SDG3) and Quality Education (SDG4).

Keywords: *Medicinal Plants, Ayurveda, Image Recognition, Artificial Intelligence, Sri Lankan Herbal Medicine*

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I wish to take this opportunity to express my sincere gratitude for the successful completion of the study proposal.

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LIST OF ABBREVIATIONS

Abbreviations	**Description**
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DB	Database
UI	User Interface
DL	Deep Learning
ML	Machine Learning
WBS	Work Breakdown Structure
SDG	Sustainable Development Goals

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1 INTRODUCTION

1.1 Background and Context

Sri Lanka is renowned for its rich biodiversity and long history of traditional Ayurvedic medicine. For centuries, Sri Lankan Ayurvedic practitioners have been using various medicinal plants, leaves, and flowers to cure illnesses and preserve human health. These plants play an important part in natural healing techniques and are frequently used in herbal medicine, home remedies, and natural treatments. However, it is difficult for the common man to identify the appropriate medicinal plant as most of the plants resemble each other and need expert knowledge to identify them.

However, with the rapid evolution of modern technology, especially smartphone application technology and artificial intelligence, it is now feasible to create intelligent systems that could aid in the identification of plants. This is because the system could use the image recognition technology to identify the plant species by taking a picture of the plant parts, such as the leaves or the flowers.

This research is designed to create a mobile-based Sri Lankan Ayurvedic plant detection system with the help of image scanning technology. With the help of this application, the user will be able to scan the leaves or flowers of the Ayurvedic plant with the camera of the mobile device. It will identify the plant and provide the user with important information regarding the Ayurvedic medicinal benefits of the plant. This will help the public become more aware of the medicinal plants found in Sri Lanka.

Sri Lanka has a long tradition of Ayurvedic medicine, which has been practiced for thousands of years. Sri Lanka is blessed with a variety of medicinal plants such as Kohomba (Neem), Gotukola, Koththamalli, Iramusu, Venivelgeta, etc. These are mostly used for Ayurvedic medicine. These medicinal plants are found abundantly everywhere. However, people are not aware of these plants without the guidance of experts on Ayurvedic medicine or botanists. Misidentification of medicinal plants may result in the improper use of the plants, which may further affect the efficiency of the Ayurvedic treatments. Further, the younger generation is gradually forgetting the traditional medicinal plants due to modernization and lack of access to reliable educational resources.

The recent advancements in the field of machine learning, computer vision, and mobile devices have enabled the development of automated plant identification systems. Deep learning techniques, such as Convolutional Neural Networks (CNNs), can be used to analyze images of plant leaves and flowers for the classification of plant species.

By using these modern technologies in conjunction with the traditional Sri Lankan Ayurvedic practices, the research proposes the development of a smart plant identification application. The application will enable users to scan images of plants and obtain information about the plant's ayurvedic uses, medicinal benefits, and treatments, thereby promoting the awareness, education, and preservation of the valuable herbal medicine of Sri Lanka.

1.2 Literature Review

Several researchers have explored ways to preserve and share herbal plant knowledge using technology.

Research 1: Ontology for Sri Lankan Medicinal Plants

Jeyananthan et al. in their research article developed an ontology for medicinal plants in Sri Lanka using the Protégé tool. The ontology has various classes such as Plants, Plant Parts, Diseases, Preparation Methods, Mixtures, and Application Categories along with their relationships. The system has covered around 200 medicinal plants and their attributes. Though the information is provided in a structured manner using ontology technology, the user has to write the query in Simple Protocol and RDF Query Language (SPARQL) in order to access the required information. But the user has to be aware of the query languages; otherwise, the system becomes difficult for the user to access the information.

Research 2: Leaf-Based Plant Identification Using Image Processing

Kumar et al. proposed a plant identification system based on leaf images by employing image processing techniques [2]. In the proposed system, the features of the leaves, such as their shape, texture, and color pattern, are extracted, and the plant species are identified by employing a classification algorithm. According to the experimental results, the proposed system can identify the different species of plants with reasonable accuracy. But the research mainly deals with the classification of plants, not specifically medicinal or Ayurvedic plants.

Moreover, the proposed system does not provide information about the medicinal benefits of the plants.

Research 3: Medicinal Plant Classification Using Convolutional Neural Networks

Singh et al. [3] proposed a method for medicinal plant classification based on a Convolutional Neural Network model. The model was trained on a dataset of images of plant leaves to automatically classify plant features. The accuracy of the model was increased when compared to traditional machine learning techniques. Even though the model is efficient in recognizing plant species, the focus of the system is on accuracy alone. There is no additional information provided on Ayurvedic properties or health benefits.

Research 4: Mobile Application for Plant Species Recognition

Silva et al. proposed a mobile application to recognize plant species by taking a picture of the plant using the camera of a mobile device [4]. Deep learning algorithms are implemented to recognize the plant species based on the image of the leaves. This method is convenient for users to recognize plant species. However, the application does not provide medicinal or Ayurvedic information related to the identified plant species.

Research 5: Deep Learning-Based Herbal Plant Recognition System

A deep learning system for the identification of herbal plants based on the images of leaves was proposed by Fernando et al. [5]. The study applied CNN to evaluate the features of the plants. The system showed promising results in classifying the plants. The study is mainly focused on the identification of the plants. The system lacks the component of the knowledge base on the use of the plants by Ayurveda.

Research 6: Automated Identification of Medicinal Plants Using Machine Learning

Patil et al. proposed an automated medicinal plant identification system using various machine learning algorithms such as SVM and image feature extraction techniques. The system identifies medicinal plant species based on the visual characteristics of the leaves. The study has proved that machine learning can be effectively employed for plant identification. However, the system is intended for research purposes and does not offer a user-friendly interface for the masses in the form of a mobile app.

Table 1-1 Summary of Existing Approaches

Approach	Strengths	Limitations
Ontology-based medicinal plant system [1]	Well-structured representation of medicinal plants, diseases, and treatment information	Need to write SPARQL queries to retrieve information; not userfriendly for non-technical users
Leaf image classification using traditional machine learning [2]	Can recognize plants by leaf features such as shape, texture, and color.	Need to perform feature extraction; less accurate compared to deep learning techniques
AI-based plant recognition platforms [3]	Fast identification and enhanced classification accuracy with deep learning techniques	Not always focused on Sri Lankan Ayurvedic plants

1.3 Research Gap

Although numerous studies have been carried out to preserve and utilize knowledge on medicinal plants of Sri Lanka, there are many unexplored areas. This subsection briefly explains the research gaps which this research study aims to address.

Gap 1: Lack of Image-Based Identification for Sri Lankan Ayurvedic Plants

The majority of the available plant identification systems are based on general plant species, making use of image recognition techniques and machine learning. However, most of these systems are based on general plant data available worldwide, but they are not focused on Ayurvedic medicinal plants specific to Sri Lanka. Therefore, it is not possible to identify local medicinal plants using these systems.

Gap 2: Limited Integration of Ayurvedic Knowledge with Plant Detection Systems

Some studies have been done on medicinal plant database systems and ontology-based systems that store structured information. However, these systems are only used to store textual

information and require specific knowledge to access the data. They don't include image recognition of plants and Ayurvedic medicinal knowledge in a simple application.

Gap 3: The Need for User-Friendly Mobile Applications to Identify Ayurvedic Plants

Several mobile applications are available to identify plants, but most of these mobile apps are designed to identify plants in general and do not offer information on the medicinal value of plants according to Ayurvedic medicine. Therefore, there is a need to develop a user-friendly mobile application to identify plants and their health benefits according to Ayurvedic medicine.

Gap 4: No Evaluation Framework for Ayurvedic Conversational AI

There is no system available to identify Sri Lankan Ayurvedic plants with image scanning and display their medicinal benefits. So, a user-friendly system is required to identify the plants and display their ayurvedic uses.

Table 1-2 Comparison of Existing Systems and Proposed System

Research Aspect	**Ontology [1]**	**General Plant Apps**	**Medicinal Plant Databases**	**Proposed System**
Image-based Plant Detection	No	Yes	No	Yes
Focus on Sri Lankan Ayurvedic Plants	Yes	No	Limited	Yes
Provides Medicinal Benefits	Yes	Limited	Yes	Yes
User-friendly Interface	Low	High	Moderate	High
Mobile Application Support	No	Yes	Limited	Yes
Integrated Plant Detection & Knowledge	No	No	No	Yes

2 OBJECTIVE

2.1 Main Objective

Sri Lanka has a wide variety of medicinal plant species, which are mainly used in Ayurveda medicine. For many generations, the knowledge of medicinal plant species, their names, their medicinal values, and their preparation has been transferred through traditional methods. However, due to modernization and the changing lifestyles of the population, the traditional knowledge of medicinal plant species is gradually being forgotten. In the present scenario, the general public cannot correctly identify the medicinal plant species and their health benefits, which has created a wide gap between traditional herbal knowledge and modern society.

In general life, people may come across various plant species in their gardens, parks, forests, and other places, but they do not have the correct knowledge about the medicinal value of the plant. When a person encounters a plant, they may think about the correct name of the plant, the correct name of the plant in Sinhala, the medicinal value of the plant, and the parts of the plant which have medicinal value. However, it is not easy to find correct information about the plant through traditional books or the internet because the information is scattered on the internet.

Thus, the main objective of this research is to develop a mobile-based system that is able to identify Sri Lankan Ayurvedic medicinal plants using image scanning technology. The proposed system will enable users to scan images of leaves or flowers of medicinal plants using a camera on a smartphone. The proposed system is able to identify the type of plant using artificial intelligence techniques. Once the plant is identified, the proposed application will provide details on the name of the plant and its Ayurvedic medicinal benefits. This proposed system aims to provide greater awareness among users regarding medicinal plants and Ayurvedic herbal medicine available in Sri Lanka.

More specifically, this research seeks to answer:

- How image recognition techniques can be used to identify Sri Lankan Ayurvedic medicinal plants using images of leaves or flowers?

- How a mobile-based application can be designed to allow users to easily and quickly identify images of leaves or flowers of medicinal plants?

- How can the system facilitate the acquisition of valuable information regarding the plant, such as its name, its medicinal benefits, and its traditional use in Ayurvedic medicine?

2.2 Specific Objectives

1. Develop an Image-Based Plant Detection Model

We want to collect, crawl, index and save the information about 10-20 of the most common medicinal plants in Sri Lanka from the web. We will gather information about the botany and taxonomy of plants including their names in different scripts (Latin, English, Sinhala), some descriptions and data about their use as medicines, including their preparation, the parts of the plant used etc. and also the diseases they are used to treat and the applications they are used for. We want to represent this knowledge base in the form of meaningful text snippets that can be embedded and searched.

2. Building a Medicinal Plant Information Database

We intend to collect data on some of the commonly used Sri Lankan Ayurvedic medicinal plants. The data is to be collected from reliable sources such as research papers, herbal medicine books, and online databases. The database will contain data on plant names, plant descriptions, medicinal benefits, parts of the plants used, and the diseases cured.

3. Develop a Mobile-Based Plant Identification Application

We plan to create a mobile-based application that will allow users to identify plants by scanning the leaves or flowers of the plants using their smartphone's camera lens. The application will send the scanned images to the model for efficient results.

4. Integrate Plant Detection With Medicinal Knowledge

We hope to incorporate the identified plant into the medicinal plant database so that the application is able to provide useful information on the identified plant, such as the name of the plant in Sinhala, the medicinal use of the plant according to Ayurveda, how the medicine is prepared, and the health benefits of the medicine.

5. Evaluate System Accuracy and User Experience

The performance of the developed system will be tested by checking the accuracy of the identification of the plants, as well as the usability of the system through the mobile application. In addition, the system will be checked to ensure it is user-friendly by using user feedback.

1. Implement Image Preprocessing Techniques

The system will also implement various image preprocessing techniques to enhance the quality of the plants' images before passing them to the plant recognition system. This is likely to enhance the accuracy of the system in identifying the plants.

2. Design a User-Friendly Mobile Interface

The system will also develop an interface that can easily be used by a mobile device user to scan the plants' leaves or flowers and retrieve the identification results. The interface will display the name, description, and medicinal value of the plants in Ayurveda medicine.

3. Evaluate System Performance and Reliability

The system will be tested with a set of images representing plants to evaluate the system's performance in identifying plants. The feedback from the users of the mobile application will also be used to evaluate the reliability of the system.

3 METHODOLOGY

Technologies In this phase, the technologies that will be used to build the Ayurvedic Plant Detection System will be explained. The technologies used will be artificial intelligence, image recognition, and mobile application technology to help a user scan a plant's leaf or flower using a smartphone's camera and get accurate identification and Ayurvedic medicinal information.

Technologies:

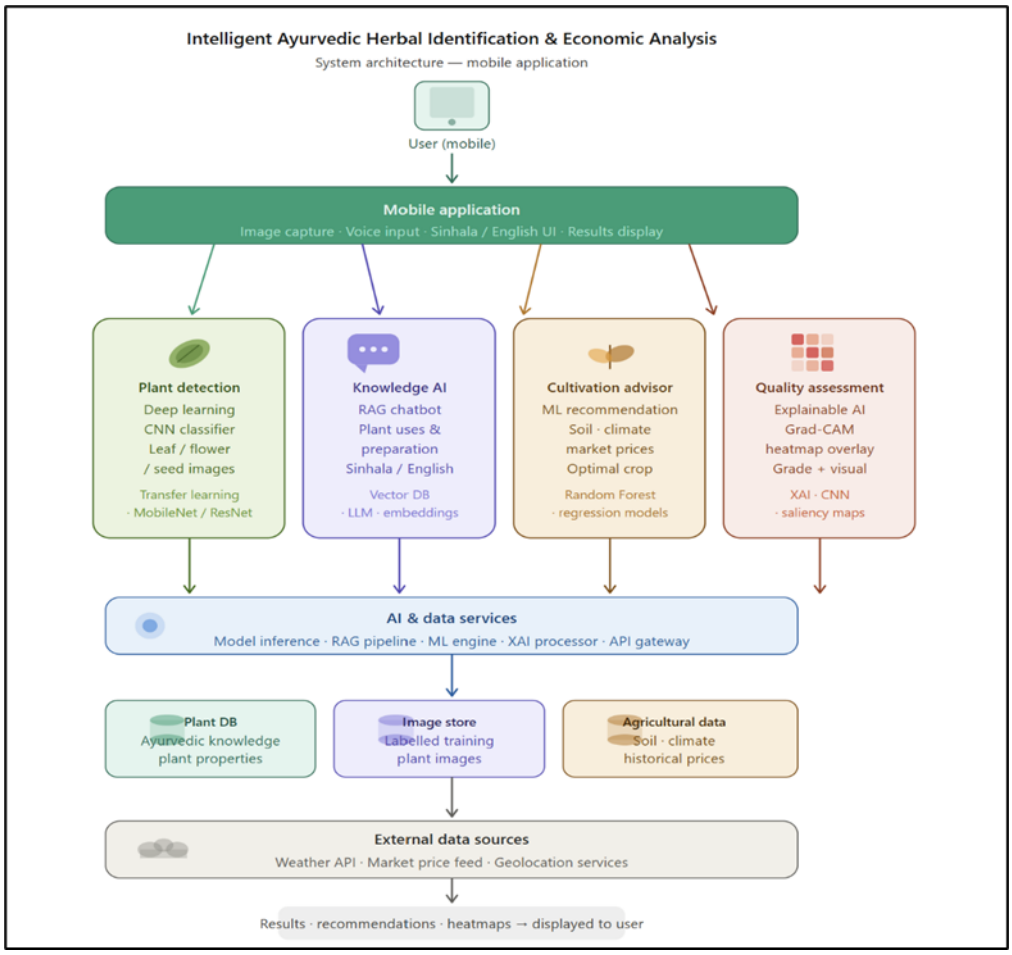
- Plant Image Dataset Collection: Images of the most common Sri Lankan Ayurvedic medicinal plants will be collected from reliable online sources. The dataset will mainly include images of leaves and flowers. In addition to the plant images, more information on the plant such as the Sinhala name, English name, description, medicinal values, plant parts, and Ayurvedic uses will also be collected.
- Image Preprocessing: The images collected of plants will undergo a process to enhance their quality. Image preprocessing includes resizing images, filtering images to remove noise, normalizing images, and arranging images into categories. This process is essential to enhance the accuracy of the plant detection model.
- Plant Identification Model: A deep learning model like Convolutional Neural Networks (CNN) will be employed to recognize the medicinal plants by analyzing the images. The images of the leaves and flowers of the plants will be used to identify the type of the plant.
- Mobile Application Development: A mobile application will be developed using Flutter or Android Studio to enable the user to capture images of the plants using the camera of the smartphone. The images will then be sent to the trained AI model to identify the plants.
- Database System: A database system is planned to be designed to hold data regarding Sri Lankan Ayurvedic medicinal plants, their names, their benefits, and their preparation methods. The application is designed to retrieve the data from the database after identifying a plant.
- Backend Framework: The backend framework will be implemented using a Pythonbased framework like Flask or FastAPI to connect the mobile application with the plant detection model and the medicinal plant database.
- Evaluation and Validation: The system developed in this project will be evaluated by using another set of data on plant images to assess the accuracy of the system. The effectiveness of the system in identifying plants will be assessed by using metrics such as accuracy and precision for the plant recognition system, and user testing for the mobile application.

Validation Strategy:

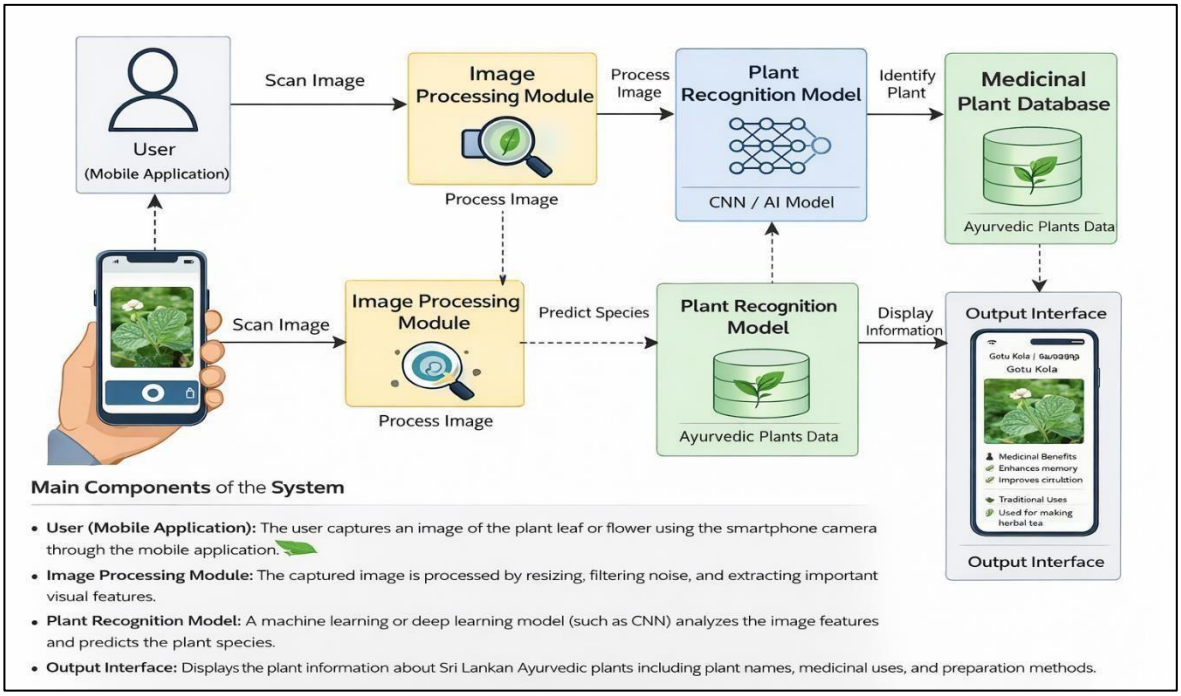
The system's performance will be evaluated using a mixed-methods approach:

- User Experience: The end users of the system, i.e., students and the general public, will be required to perform the usability test to assess the ease of use of the system. The responses of the users will be recorded with regard to the user interface of the system, the scanning of the plants, and the overall satisfaction of the users.
- Plant Identification Accuracy: The performance of the system with regard to the identification of the plants will be checked by employing the test dataset consisting of the images of the leaves and flowers of the plants. The accuracy of the system with regard to the identification of the Sri Lankan Ayurvedic medicinal plants will be checked.
- Information Quality: The information with regard to the medicinal properties of the identified plants will be retrieved by the system. The accuracy of the system with regard to the information retrieved will be checked by referring to authentic Ayurvedic literature, if possible, with the approval of Ayurvedic experts.

4 HIGH-LEVEL SYSTEM ARCHITECTURE



4-1 System Overview Diagram



4-2 Component Overview Diagram

User Query Flow:

- The user opens the mobile application.
- The picture of a plant's leaves or flowers with the camera of the mobile phone.
- The captured image is sent to the image processing module
- The system processes the image.
- The processed image is sent to the plant recognition module.
- The system analyzes the image and identifies the plant.
- The system retrieves the Ayurvedic medicinal information related to the plant.
- The application shows the user the plant's name, medicinal value, and uses.

5 USER REQUIREMENTS

5.1 Stakeholder Identification

- General Users: This group consists of the users of the mobile application who are interested in identifying Sri Lankan Ayurvedic medicinal plants and their medicinal benefits.

- Ayurvedic Practitioners: This group consists of traditional medicine practitioners who are interested in utilizing the Ayurvedic Plant Detection System.
- System Developers: This group consists of the individuals who are responsible for the development of the mobile application.
- Health & Herbal Medicine Communities: This group consists of communities that are interested in the development & utilization of Ayurvedic & traditional herbal medicine knowledge.

5.2 Functional Requirements

- FR-01 The system should enable users to capture the plant leaves or flowers' images through a smartphone's camera.
- FR-02 The system should process the captured image to identify the plant species through image recognition technology.
- FR-03 The system should display the identified plant's name.
- FR-04 The system should display the Ayurvedic medicinal information related to the identified plant.
- FR-05 The system should display the plant's description, medicinal value, and plant parts' use in treatment.

5.3 Non - Functional Requirements

- Performance and Efficiency:
  The system must be able to recognize the scanned image of the plant within a reasonable time.

The system shall be able to support multiple users using the application simultaneously. The system shall be able to provide accurate identification of plants to most test images. • Usability:

The system must be able to provide a simple and user-friendly interface to scan plants.

- Scalability:
  The system should be able to add new medicinal plants to its database easily.
- Security and Privacy:
  The system should be able to protect its users and prevent unauthorized usage of its database.
- Accuracy:
  The system should be able to classify the test images of the plants accurately. The system should be able to provide accurate Ayurvedic medicinal information on the identified plants.

6 COMMERCIALIZATION PLAN

The Ayurvedic Plant Detection System is targeted towards the Sri Lankan public, students, and interested parties who are interested in the field of herbal medicine. The mobile app will have the capability to detect the leaves or flowers of plants, giving the user information about the plants, their Ayurvedic, and medicinal uses. The system can be monetized by providing a freemium service, where the basic service will be free, and the advanced service will be premium. It can also be licensed to educational and research institutions interested in the field of medicinal plants.

Revenue Model:

Table 6-1 Revenue Table

Model	**Description**	**Estimated Revenue**
Freemium App	Basic plant identification is free; premium features provide detailed medicinal information and advanced features	LKR 50,000/year (100 users)

Institutional Licensing	Hospitals/ Universities (LKR 50,000/Year)	LKR 300,000/Year (10 institutions)
Data Insights	Anonymous plant usage data can be used for research and environmental studies	LKR 100,000/Year
Total Potential Annual Revenue		**LKR 650,000/Year**

Cost Estimation: The cost of maintaining the system includes cloud hosting services, model training infrastructure, application maintenance, and system updates.

Table 6-2 Cost Estimation Table

Cost Category	**Initial Development**	**Annual Operating**
Cloud Hosting	LKR 20,000	LKR 60,000
AI Model Services	LKR 10,000	LKR 40,000
Maintenance	-	LKR 80,000
Total	**LKR 30,000**	**LKR 180,00**

7 BUDGET AND JUSTIFICATION

Table 7-1 Estimated Budget Table

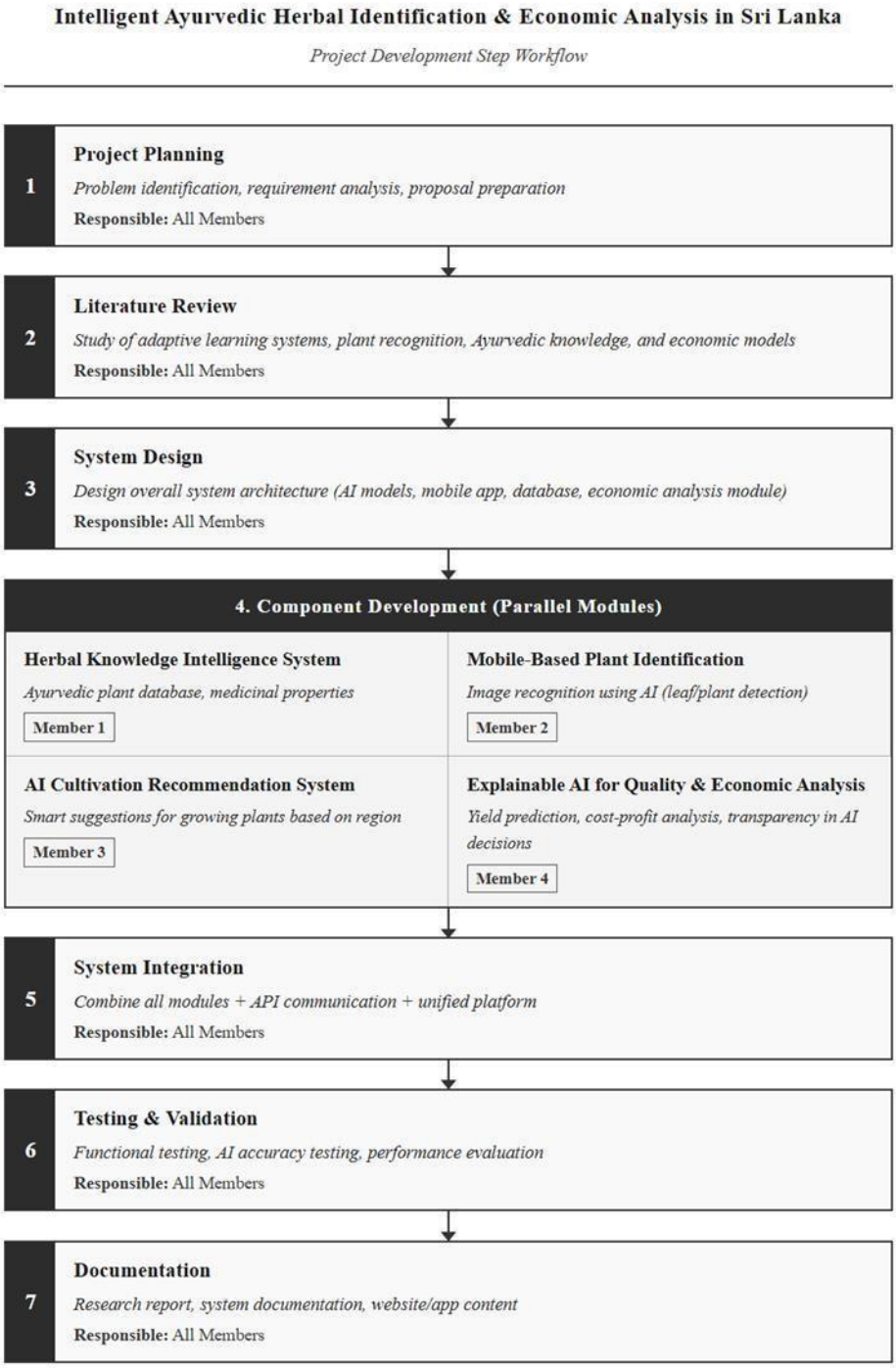
Description	**Total Amount (Rs.)**
Software and Tools (AI libraries, development tools, mobile app IDE)	10,000.00
Cloud Computing and Hosting	12,000.00

Dataset Collection and Image Resources	15,000.00
Transportation and Communication	5000.00
Printing and Documentation	3000.00
Paper Publishing	10,000.00
Total	**48,000.00**

Budget Justification:

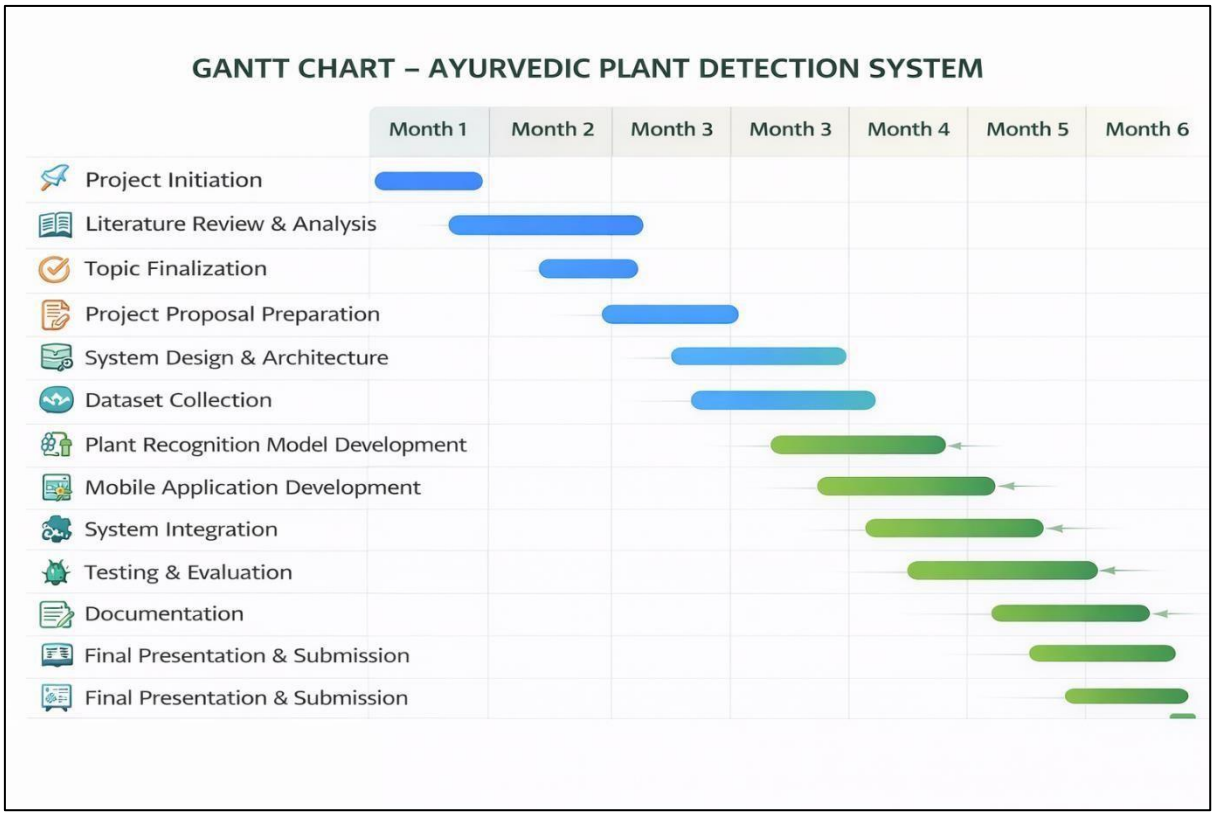
- **Software and Tools (LKR 10,000): This category includes software development tools, AI software frameworks, and possible API services required to develop the plant detection system and mobile application.**
- **Cloud Computing and Hosting (LKR 12,000):** This software is used to host the backend server and run the machine learning model to identify the plant.
- **Dataset Collection and Image Resources (LKR 8,000):** This is the amount required to collect plant images and prepare a dataset of Sri Lankan Ayurvedic medicinal plant varieties.
- **Paper Publishing (LKR 10,000):** If the research findings are to be published in a conference or journal, the publication costs may be incurred.

8 WORK BREAKDOWN STRUCTURE (WBS)



8-1 Work Breakdown Structure

9 GANTT CHART



9-1 Gantt Chart

10 REFERENCE LIST

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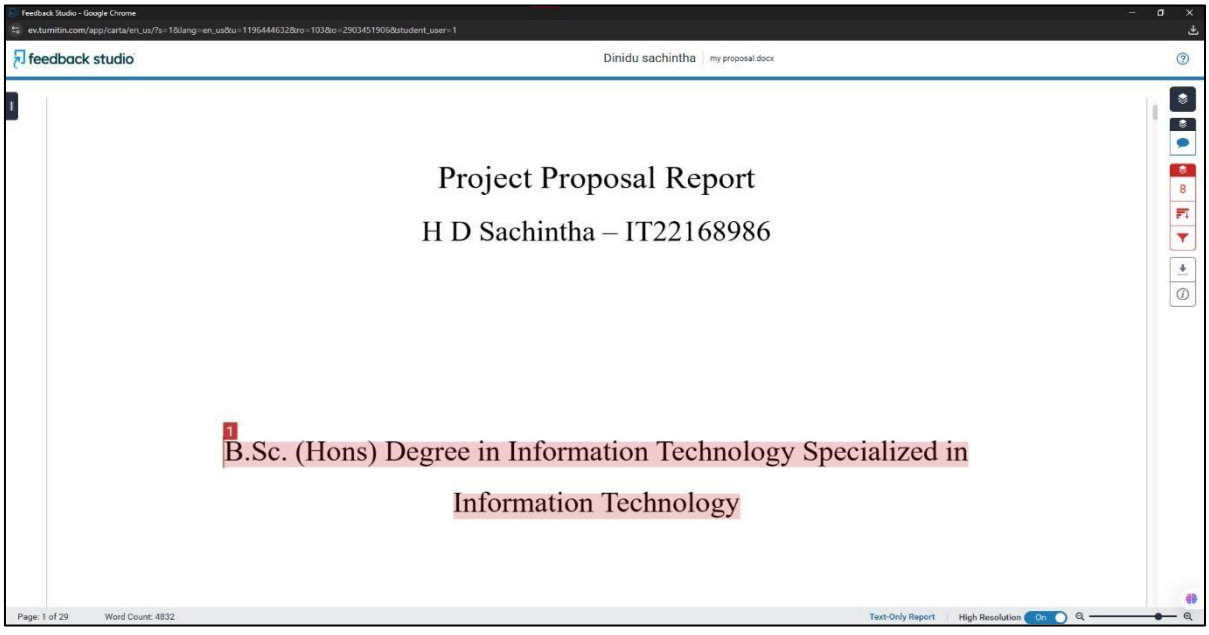
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11 APPENDICES



Appendix 1- Plagiarism Report